The constant in [OpenAI, *Uniform exclusion of Landau–Siegel zeros*] is effective: $c \geq 3.39 \times 10^{-3}$

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Computation: `src/effectivity.py`; output: `results/2026-10-20-effectivity/effectivity-output-v2.txt`. Derivation: `docs/effectivity/derivation.md`. Sources and quotations: `docs/effectivity/sources.md` (S1–S8). This PDF is built from `docs/effectivity/note.tex`, which follows `note.md`.

1 Statement

Assuming the proof in [S1] is correct (see §7), every real zero $\beta \in (0,1)$ of every primitive nonprincipal real Dirichlet L-function of conductor $q \geq 3$ satisfies

$$(1-\beta)\log q > 3.399 \times 10^{-3} \quad \text{using Platt's computation for } q \leq 4 \cdot 10^5,$$

$$(1-\beta)\log q > 4.467 \times 10^{-3} \quad \text{using Lu–Zaman–Zhao's computation for } q \leq 10^{10} \text{ instead.}$$

The argument of [S1] alone, without any computation for small q, gives $(1-\beta)\log q > 5.04 \times 10^{-4}$ for every $q \geq 3$ except $q = 8$. These bounds are certified in Arb ball arithmetic from exact rational parameters.

2 What the paper proves and what is added here

- **The paper (Theorem 1):** some absolute $c > 0$ exists. Its proof argues by contradiction along a sequence with $q \to \infty$ and $\delta := (1-\beta)\log q \to 0$, and it leaves every analytic constant implicit ($\ll$, C, C_H).
- **Added here:**
  - every implicit constant replaced by an explicit one, with a cited source and its range of validity;
  - the limiting argument replaced by a direct inequality valid for each $q \geq q_0$;
  - the parameters (γ, H) chosen to maximize the resulting bound;
  - a verification of every side condition on which the paper's lemmas depend.

No step of the paper's algebra (Lemmas 3, 6 and 7, Corollary 4) is changed.

3 Explicit inputs

Input	Statement used	Source and range
Hadamard identity (2.3)	$\mathrm{Re}(-L'/L)(s,\chi) = \frac{1}{2}\log(q/\pi) + \frac{1}{2}\psi((s+\varepsilon)/2) - \sum_\rho \mathrm{Re}\,\frac{1}{s-\rho}$, terms ≥ 0, real $s > 1$	as stated in [S1] §2; checked against Davenport, 2nd ed., Ch. 12 (17)–(18), p. 83
ζ log-derivative	$-\zeta'/\zeta(\sigma) < 1/(\sigma-1)$, $\sigma > 1$	[S6] Lemma 12; its final elementary inequality proved here for all $\sigma > 1$ (§4)
Digamma	$\psi((s+\varepsilon)/2) \leq \psi(3/2) = 2 - \gamma - 2\log 2$ for $1 < s \leq 2$	ψ increasing
Mertens	$\log x + E - \frac{1}{2\log x} < \sum_{p\leq x} \frac{\log p}{p}$ $(x > 1)$; $\sum_{p\leq x} \frac{\log p}{p} < \log x + E + \frac{1}{2\log x}$ $(x \geq 319)$	Rosser–Schoenfeld Thm 6, (3.21)–(3.22), p. 70, read in the original
Chebyshev	$\theta(x) < (1 + 7.5\cdot 10^{-7})x$, $x > 0$	[S5] Platt–Trudgian Cor. 2
Primes dividing q	$\sum_{p\mid q} \frac{\log p}{p} \leq \min\left(\frac{1}{2}, \frac{\log 2}{2\log q} + \frac{1}{3}\right)\log q$	elementary
Lemma 6	$S_2 \leq 192\, MH^{-1/3}U$, $S_1 \geq MH^{2/3}U/(4\cdot 97^2)$, given $N \geq H$ and $H^{2/3}N^{4/3} \geq 2\cdot 97^2$	[S1] lines 517–550
Small q	no real zero with $\sigma > 0$ for $q \leq 4\cdot 10^5$ (Platt 2016); $\delta > 1/5$ for $q \leq 10^{10}$	Platt 2016 Thms 10.1–10.2, read in the original; [S2] Theorem 1.1

4 The direct inequality and the parameter choice

Combining the explicit Lemma 2, (5.1), (5.6) with the θ bound, and Lemma 6 (derivation.md §§1–5): if a real zero with $\delta = (1-\beta)\log q$ exists and $N \geq H$, then

$$1 \leq \tfrac{3}{4}(1+\rho_H) + \frac{C_\ell + \frac{1}{2}(1+\rho_H)}{\lambda} + \frac{e}{2}\,\delta\lambda + \frac{C_H + (1+\rho_H)\log 8}{L} + \frac{(1+7.5\cdot 10^{-7})\cdot 4\cdot 97^2}{H^{2/3}L} + \frac{6\cdot 97^2}{H^{2/3}U},$$

where

- $\rho_H = 7{,}226{,}112/H$, $C_\ell = e/4 + c_q$, $C_H = \log H + \frac{1}{2\log H} + \frac{1}{2L} + \frac{e}{2}\kappa$;
- $\kappa = \frac{1}{2}\psi(3/2) - \frac{1}{2}\log\pi = -0.55412\ldots < 0$;
- $L = \log U = \frac{4}{3}\log N$ and $\lambda = L/\log q$.

Take $N = \lceil q^\gamma \rceil$. For all $q \geq q_0$:

- λ lies in $[4\gamma/3,\ 4\gamma/3 + \frac{4}{3}\log(1+q_0^{-\gamma})/\log q_0]$;
- every term except the δ term is largest at $q = q_0$;
- so every real zero with $q \geq q_0$ has $\delta > \delta_0(q_0, \gamma, H)$, with δ_0 given in closed form (function `delta0` in `src/effectivity.py`).

The paper's split of the margin $(13/16 + 1/16)$ is not needed: for fixed (γ, H), δ_0 is determined exactly.

The parameters were found by a floating-point search and then certified in Arb at the exact values below (`effectivity-output-v2.txt`; every printed bound is the Arb lower endpoint rounded down).

Regime	q_0	γ	H	c_q	certified $\delta_0 >$
Platt	400,001	201/10	1,995,262,314	$\frac{\log 2}{2\ell} + \frac{1}{3}$	$\mathbf{3.3992904026 \times 10^{-3}}$
Lu–Zaman–Zhao	$10^{10}+1$	153/10	2,818,382,931	$\frac{\log 2}{2\ell} + \frac{1}{3}$	$\mathbf{4.4673780338 \times 10^{-3}}$
argument alone	3	671/5	1,122,018,454	$\frac{1}{2}$	$\mathbf{5.0425352961 \times 10^{-4}}$

At the paper's own $H = 86{,}713{,}344$, the same three regimes give 2.064, 2.652 and 0.3245 ($\times 10^{-3}$). Optimizing over H is worth a factor of about 1.6.

Side conditions (derivation.md §7), certified at q_0 and implied for all $q \geq q_0$: $H \geq 319$; $N \geq H$ ($\gamma \log q_0 > \log H$); $H^{2/3}N^{4/3} \geq 2 \cdot 97^2$ (from $N \geq H \geq 319$); $U \geq 3$, and $s = 1 + 1/L \in (1, 2]$; $\kappa < 0$.

Lemma 12's last step. $\varphi(\sigma) = \frac{1}{2\sigma-1} - \left[\frac{1}{\sigma-1} - \frac{1-2^{-\sigma}}{\sigma} - \frac{3^{1-\sigma}}{\sigma-1}\right] > 0$ for all $\sigma > 1$. Proved in three pieces:

- on $(1, 1.01]$, by the exact chain $(1-3^{-t})/t \leq \log 3$ and $(1-2^{-\sigma})/\sigma \geq 1/(2\sigma)$, which reduces it to $\log 3 - 1/2.02 < 1/1.02$;
- on $[1.01, 5]$, by Arb on 4,000 intervals (minimum certified lower bound of φ: 0.03878);
- for $\sigma \geq 5$, by $\varphi > 0 \Leftarrow \sigma^2 - 5\sigma + 2 > 0$.

5 Small conductors

- **Platt regime:** for $q \leq 4 \cdot 10^5$ there is no real zero with $\sigma > 0$ at all, so the bound holds vacuously. This range includes $q = 8$, the conductor the paper's field construction excludes ($d = 2$). $q > 4 \cdot 10^5$ is covered by the first row.
- **Lu–Zaman–Zhao regime:** for $q \leq 10^{10}$, $\delta > 1/5$, which also covers $q = 8$; $q > 10^{10}$ is covered by the second row; so $c = \min(4.467 \cdot 10^{-3}, 1/5) = 4.467 \cdot 10^{-3}$.
- **Argument alone:** covers $q \geq 3$ with $q \neq 8$ by taking γ large ($\lambda \approx 179$), i.e. $N \approx q^{134}$. $q = 8$ needs a separate input.

6 The final constant

$$c = 3.399 \times 10^{-3} \text{ (with Platt)}, \qquad c = 4.467 \times 10^{-3} \text{ (with Lu–Zaman–Zhao)},$$

both rounded down from the certified values.

Context.

- The companion paper on the zero-free half-plane $\operatorname{Re} s > 7/8$ would, if correct, give $1 - \beta \geq 1/8$ for every such β, a far stronger statement.
- The point here is narrower: the nine-page Landau–Siegel argument, on its own, yields an explicit constant.
- A real zero with $\delta \approx 0.003$ is far outside anything ever observed. These numbers are for checking the argument, not a claim about zeros.

Comparison with an earlier version (docs/explicit-constant.md, 2026-10-08), whose figures were $3.0379 \cdot 10^{-4}$, $1.99806 \cdot 10^{-3}$ and $2.58930 \cdot 10^{-3}$:

- **Same inputs reproduce it.** With its inputs switched back on, this code gives slightly larger maxima ($3.040 \cdot 10^{-4}$, $2.0000 \cdot 10^{-3}$, $2.5918 \cdot 10^{-3}$, rounded down). The earlier note reported a δ with a positive safety margin ($F \approx 9.4 \cdot 10^{-5}$) rather than the maximum.

- **Its "all $q \geq 3$" figure** covered small q the same way as row 3 here: by taking λ large, not by any computation ($q = 8$ again excepted).
- **It used three inputs replaced here:** an unsourced "$\sum_{p \leq x} \log p/p < \log x$" for the primes up to H, which costs $|E| = 1.33$ in C_H against the sourced Rosser–Schoenfeld upper bound; Rosser–Schoenfeld Thm 9 for θ; a fixed H.
- **The gain comes mainly from freeing** H ($\times 1.6$). Of the other changes, $c_q = \frac{1}{2}$ is worse than its c_q at large q but better at $q = 3$, and the sourced Mertens bound and the θ bound are improvements.

7 Assumptions (everything the constant depends on)

1. **The paper's proof is correct:** in particular Lemma 3, Corollary 4, Lemma 6, Lemma 7 and the bounds (5.1) and (5.6).
   - This note does not re-verify them mathematically.
   - Independent evidence: the Lean formalization of Theorem 1 is accepted by the official comparator with two kernels (Lean and nanoda), including a version rerouted through the paper's own Lemma 3. See the replay packet (DOI 10.5281/zenodo.23269527) and the statement audit on `grwtsk/openai-math` #40.
   - The Lean proof has existential constants only and does not contain these numbers.
2. **The Hadamard identity (2.3).** Checked against Davenport, *Multiplicative Number Theory*, 2nd ed., Ch. 12, (17) and (18) on p. 83, with $a = \varepsilon$ (p. 70). Taking real parts of (17) and using (18), $\operatorname{Re} B(\chi) = -\sum \operatorname{Re} 1/\rho$, gives (2.3).
3. **Rosser–Schoenfeld Theorem 6,** read in the original: (3.21) for $1 < x$ and (3.22) for $319 \leq x$, p. 70. $E \approx -1.33258\,22757\,33221$, (2.11). This agrees with the quotation in arXiv:1703.08032 and with the enclosure in `src/effectivity.py`.
4. **Platt (2016),** read in the original. Theorem 10.1 states GRH for primitive characters of modulus $q \leq 400\,000$, up to a positive height. Theorem 10.2 states $L(1/2, \chi) \neq 0$ for $q \leq 2\,000\,000$. Together (our inference) they rule out a real zero in $(0, 1)$ for $q \leq 4 \cdot 10^5$, matching the Lu–Zaman–Zhao summary. Needed only for the $3.399 \cdot 10^{-3}$ statement.
5. **Lu–Zaman–Zhao Theorem 1.1** (arXiv:2602.03626v1, a preprint), read in the original. Needed only for the $4.467 \cdot 10^{-3}$ statement.
6. **Platt–Trudgian Corollary 2,** read in the original.
7. **Lemma 12 of arXiv:2511.13898** (a preprint), read in the original. Its reduction steps are taken from that proof; its final inequality is proved here.
8. $\chi(p) = (d/p)$ **for odd** $p \nmid q$**, and** $|d| \leq q$**,** as stated in the paper.

Not used: any unproved hypothesis (GRH or similar), or any floating-point value in a claimed inequality.

References

[S1] OpenAI, *Uniform exclusion of Landau–Siegel zeros*, October 1, 2026, `github.com/openai/math`, commit adc7f12, family 003.

[S2] R. F. Lu, A. Zaman, H. Zhao, *Numerical computations concerning Landau–Siegel zeros*, arXiv:2602.03626v1 (2026).

[S3] J. B. Rosser, L. Schoenfeld, *Approximate formulas for some functions of prime numbers*, Illinois J. Math. 6 (1962), 64–94, `doi:10.1215/ijm/1255631807`.

[S4] Axler, *New estimates for some functions defined over primes*, arXiv:1703.08032.

[S5] Platt, Trudgian, *On the first sign change of* $\theta(x) - x$, arXiv:1407.1914.
[S6] Upper bounds on gaps between zeros of L-functions, arXiv:2511.13898, Appendix, Lemma 12.
[S7] H. Davenport, *Multiplicative Number Theory*, 2nd ed., rev. H. L. Montgomery, Springer GTM 74.
Also: D. J. Platt, *Numerical computations concerning the GRH*, Math. Comp. 85 (2016), 3009–3027, doi:10.1090/mcom/3077.